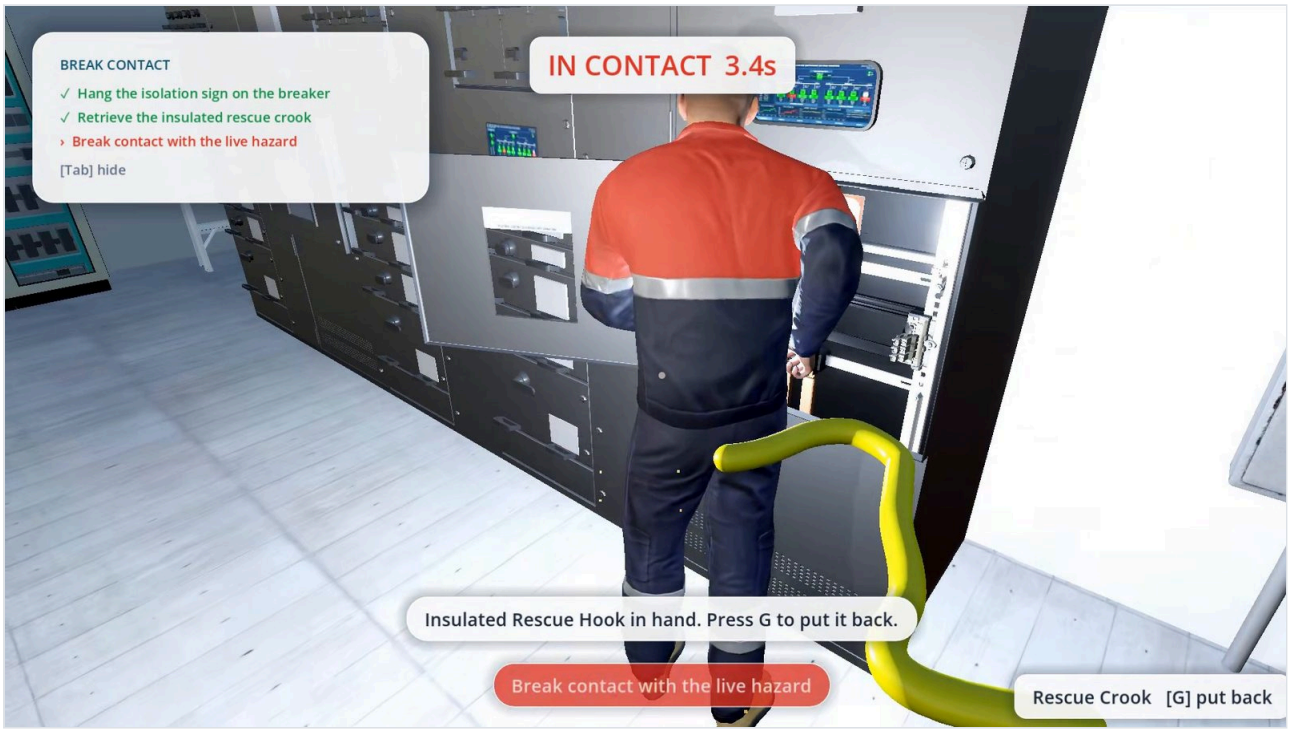
SERVICE OFFER

Custom training simulators

I build 3D simulators in which the learner carries out your procedure, every step is graded against it, and the result is recorded in your training platform (LMS).



Low-voltage rescue module: the worker is in contact with the live board, a clock counts the contact time, and the learner holds the insulated rescue hook.

Nadir Zouaoui

Electrical engineer / Training simulator developer

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Nadir Zouaoui, electrical engineer. Based in Oran, Algeria. Registered auto-entrepreneur (auto-entrepreneur card), NIF 198220102511152022 80. Profile: https://anae.dz/id/9DDvcMwes9E4RNFHsRLV

What I build

Procedure trainers. The learner carries out the task in a first-person 3D scene. Every step is graded against the procedure, and the result is reported to the organisation's training platform, the LMS.

For training organisations and for companies that train their own staff: utilities, mining, energy and industry. The simulator is built around your procedures.

What is an LMS?

An LMS (Learning Management System) is an organisation's online training platform. It makes the modules available to staff and keeps, for each person, the modules taken and the results obtained. The simulator is loaded into it as a module and records its results there.

If you do not have an LMS, I can supply one, customised for the project.

Any scenario

Any procedure you can document, built from your own documents: switching programmes, rescue procedures, permits, site rules, equipment-specific tasks.

Any platform, any graphics level

Web browser, Windows desktop, tablet, or VR/AR headset. The two examples in this document run in a browser, so their graphics are deliberately light to load on any office PC. Desktop and VR builds can use more advanced render engines.

Modular

Modules are assembled from reusable parts. A scenario can be delivered whole or as shorter standalone modules. Example: the CPR part of the low-voltage rescue scenario also exists as a standalone module.

Built by an electrical engineer

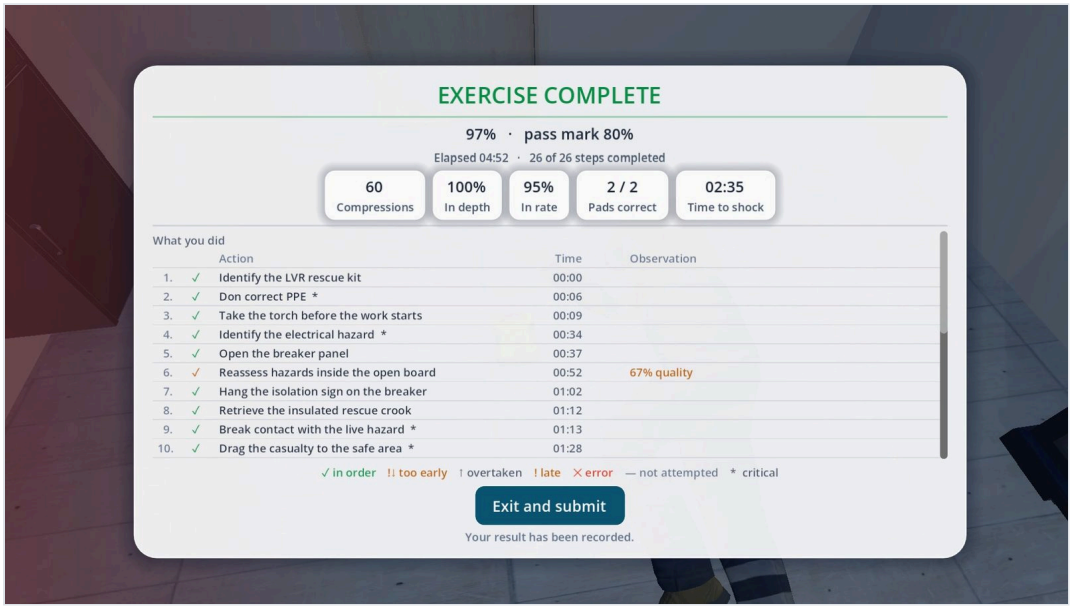
State engineer in electrical engineering, four years of experience in oil and gas, industrial automation and renewable energy, certified for electrical equipment in hazardous areas (EEHA, AS/NZS 60079). I model the scenes in Blender and build the simulators in Godot 4 myself, so the technical content is right, not only the visuals.

THE CORE OF THE SERVICE

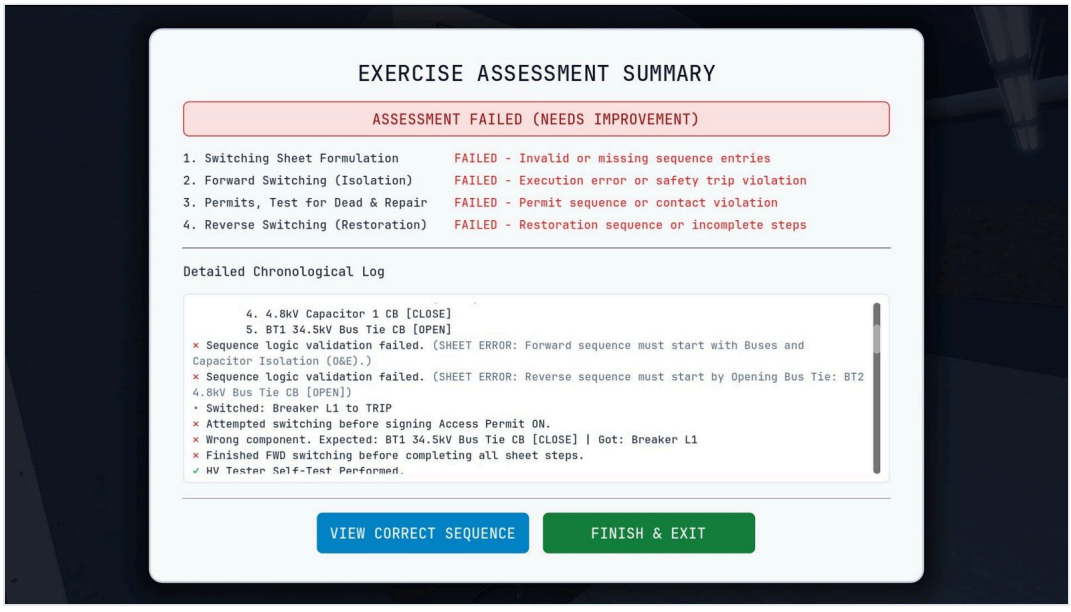
Grading and results reporting

Each step is graded against your procedure, with the criteria you set.

- Weights, critical steps and pass mark are yours. The order and the timing of the steps are assessed as well.
- A wrong choice is recorded, not blocked. The simulator lets the learner make the error and logs it.
- Status, score and the step-by-step log are sent to your LMS through SCORM, the standard exchange format between a training module and an LMS.



Low-voltage rescue, results screen: 97 % against a pass mark of 80 %, 26 of 26 steps, compressions in depth and in rate, pads placed, time to shock, then the time of each step.

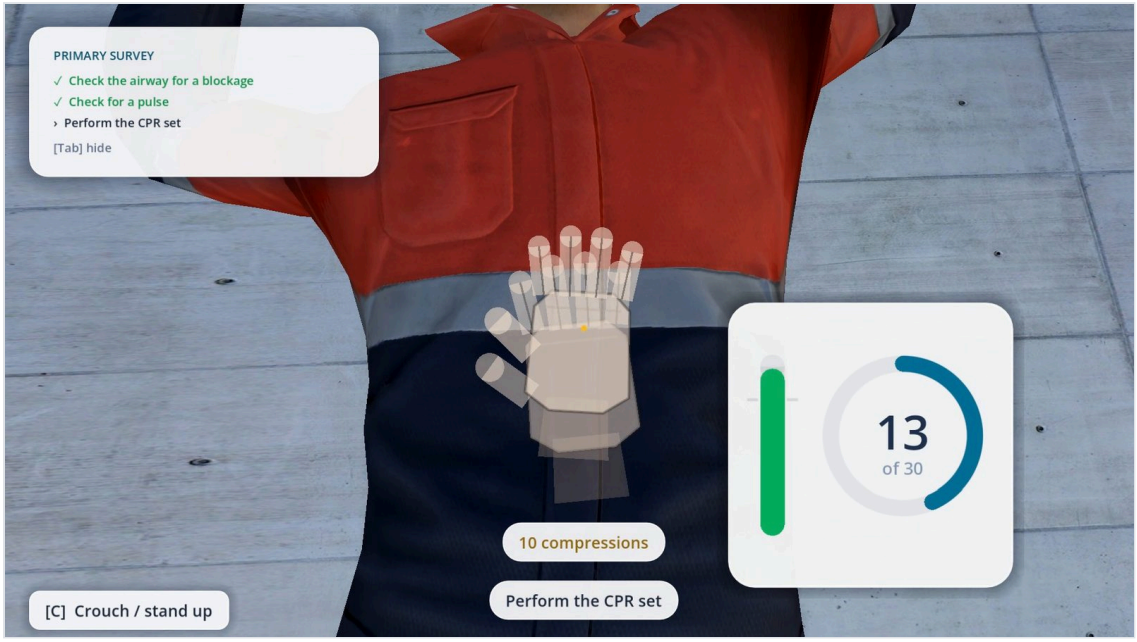


HV switching, a failed attempt: each of the four parts carries its reason, and the chronological log names every error.

EXAMPLE 1

Low-voltage rescue and CPR

Made for an Australian registered training organisation, which uses it to assess low-voltage rescue and CPR competency. Runs in the browser, inside the LMS.



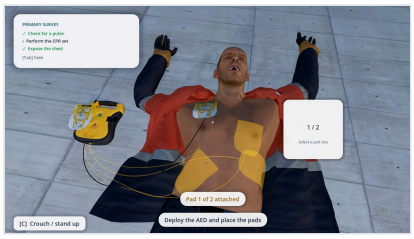
Chest compressions: counter at 13 of 30 and depth bar. Each compression is measured as it is given.



Rescue kit check: each item is selected on the bench and named from a menu.



After contact is broken: drag the casualty to the safe area and isolate the circuit at the breaker.



Defibrillator: first pad attached, the second still to be placed on the chest.

What the learner does

A worker opens a live low-voltage board and receives an electric shock. Before that, the learner checks and names the rescue kit, identifies the hazards and marks the isolation point. When the shock happens, the learner breaks contact with the insulated hook, drags the casualty clear, isolates the supply and calls for help, then carries out CPR and uses the defibrillator until the ambulance arrives.

What the LMS receives

Completion status, score and the step record, through SCORM 1.2. A failed attempt is recorded as well.

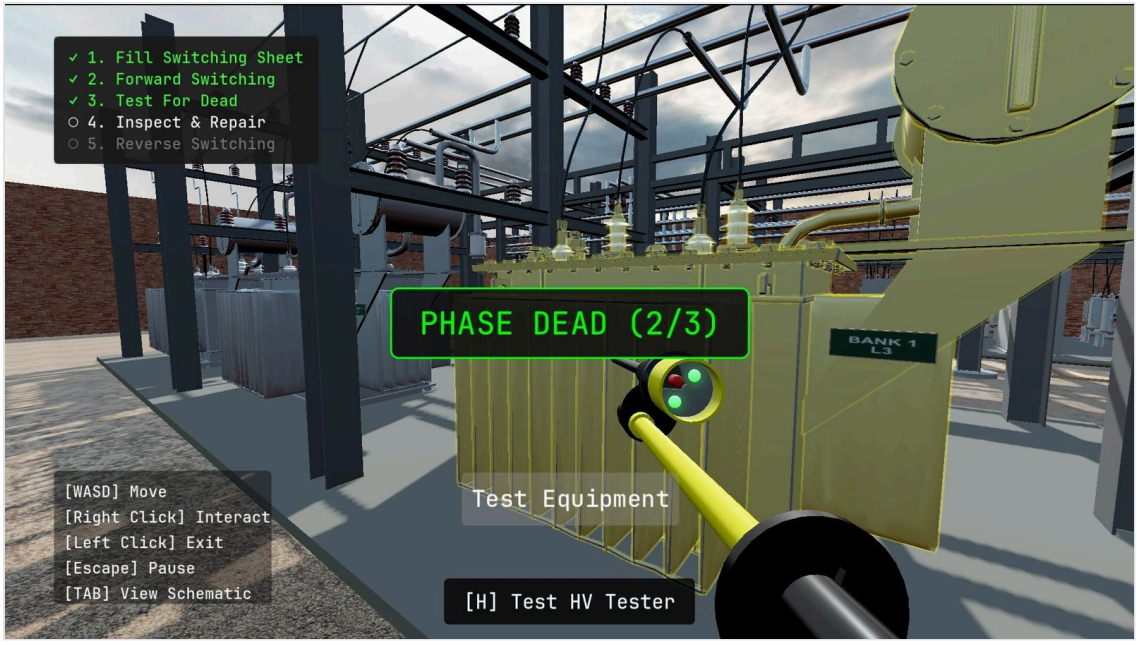
How it is graded

1. 26 steps are graded, each with its own weight. The pass mark is 80 %.
2. A step done out of order or late keeps only part of its marks.
3. Seven steps are critical: getting one wrong fails the attempt whatever the score.
4. A fatal safety error, such as touching the casualty bare-handed while still in contact, ends the exercise at once.
5. Compressions are measured one by one for depth and rate, and the time to the first shock is recorded.

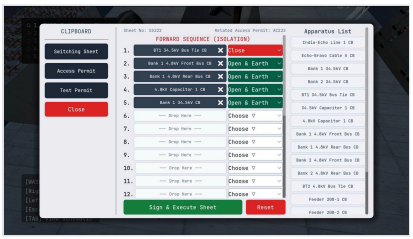
EXAMPLE 2

HV substation switching

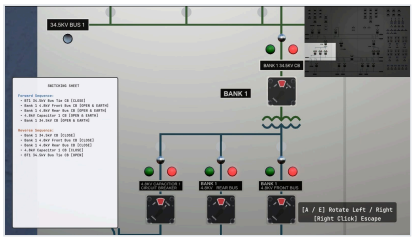
Made for an Australian registered training organisation, which uses it to assess HV switching competency. The substation and its diagram are a training layout, not a real site.



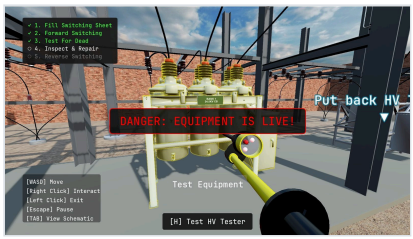
Proving dead in the yard with the HV tester: second phase of three dead.



Switching sheet: each apparatus is dragged into a numbered step and given its operation.



Mimic panel in the control room, with the learner's own sheet beside it.



HV tester held at a breaker that is still live: the warning shows in red.

What the learner does

A transformer bank in a 34.5 kV / 4.8 kV substation has a fault. The learner writes the switching sheet, isolates and earths the bank from the control room, signs on to the access permit and the test permit, proves the transformer dead in the yard, repairs it, then restores supply in the reverse order.

What the LMS receives

Completion status, score and the chronological log, through SCORM 1.2. The module runs inside the LMS with nothing to install.

How it is graded

1. Four parts are assessed separately: writing the switching sheet, forward switching (isolation), permits with test for dead and repair, and reverse switching (restoration).
2. The sheet written by the learner is checked against the sequence rules. A wrong order is named in the log.
3. The log records every operation in time order, including operating the wrong apparatus, switching before the access permit is signed on, and steps left out.
4. After the attempt, the correct sequence is shown with the purpose of each step.

How a project runs

Six steps, each with a part for you. The scene and the interactions are built in two stages: you give your feedback on the scene before it is frozen.

1

Your requirement

You send the procedure as you apply it, your pass criteria and the documents available on the installation: diagrams, photos, switching sheets, site rules. We settle the platform, the language and the graphics level.

YOU RECEIVE
A quote and a schedule drawn up for your scenario.

2

Scenario and grading specification

I write the scenario step by step, then the grading specification: weight of each step, critical steps, pass mark, order and timing rules, errors to record.

YOU APPROVE
Both documents, before the build starts.

3

3D scene build

I model the site, the equipment and the tools. At this stage the scene can be walked through but has no interactions yet. You compare it with your installation.

YOU REVIEW
The scene. Your comments are integrated, then it is frozen.

SCENE FROZEN

4

Interactions and scoring

On the frozen scene, I program the steps of the procedure, the error cases, the scoring, the results screen and the reporting to the LMS.

YOU TEST
A complete version, and report the corrections needed.

5

Tests in your LMS

The module is loaded in your LMS, or in the one I supply. Status, score and step log are checked there, on a passed attempt and on a failed one.

YOU CHECK
That the results arrive in your platform.

6

Delivery

The SCORM package ready to load, or the build for the chosen platform: Windows, tablet or headset.

YOU RECEIVE
The module, ready to release to your learners.

Platforms

PLATFORM	HOW IT IS DELIVERED	WHAT IT SUITS
Web browser (SCORM)	SCORM 1.2 module that runs inside the LMS, with nothing to install. Graphics kept light to load on any office PC. Both examples in this document.	Learners who take the module from the LMS on an office PC.
Windows desktop	Windows application. Can use more advanced render engines. Both examples also run on Windows.	Scenes that need higher visual fidelity than a browser build.
Tablet	Touch-screen build.	Sessions in a training room or on site, without a PC. Touch interactions.
VR/AR headset	Headset build. Can use more advanced render engines for higher visual fidelity.	Procedures where movement and body position matter, carried out at real scale.

- Outside the browser (Windows, tablet, headset), results are sent to the LMS over a network connection, for example through xAPI, or exported to a file. The reporting method is set in the specification.
- If you do not have an LMS, I can supply one, customised for the project.
- Simulators can be delivered in English or in French.

Request a quote and a schedule for your project

Send me your procedure, or simply describe what you need. I reply with a quote and a schedule drawn up for your scenario.

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WHATSAPP +213 656 060 214

WEBSITE https://nadirzouaoui.github.io/simulators/

LINKEDIN linkedin.com/in/nadir-zouaoui

A live demo of the two modules is available on request. Short clips are on the website.